Project 4 Report

CECS 327 - Intro to Networks and Distributed Computing

Video: https://youtu.be/Wp4sNzWtgGc

GitHub: https://github.com/harvest7777/project4

Due: May 2nd, 2026

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System Design

Flask Node (node.py)

Every request goes through here and potentially gets routed to a different node. All nodes share the same source code.

Storage Volume (/app/storage)

Each node gets its own local directory to store uploaded files. We mount it as a Docker volume so the files persist.

Key-Value Store (store dict)

In memory kv store. It only holds the keys that the DHT decided this node is responsible for. We use a hashing algorithm to figure out which node to route to.

DHT Router (responsible_node)

We SHA-1 hash the key and use that to figure out which node owns it. If the current node isn't responsible, it forwards the request to whoever is responsible.

EXTRA CREDIT: Health Monitor (health_check thread)

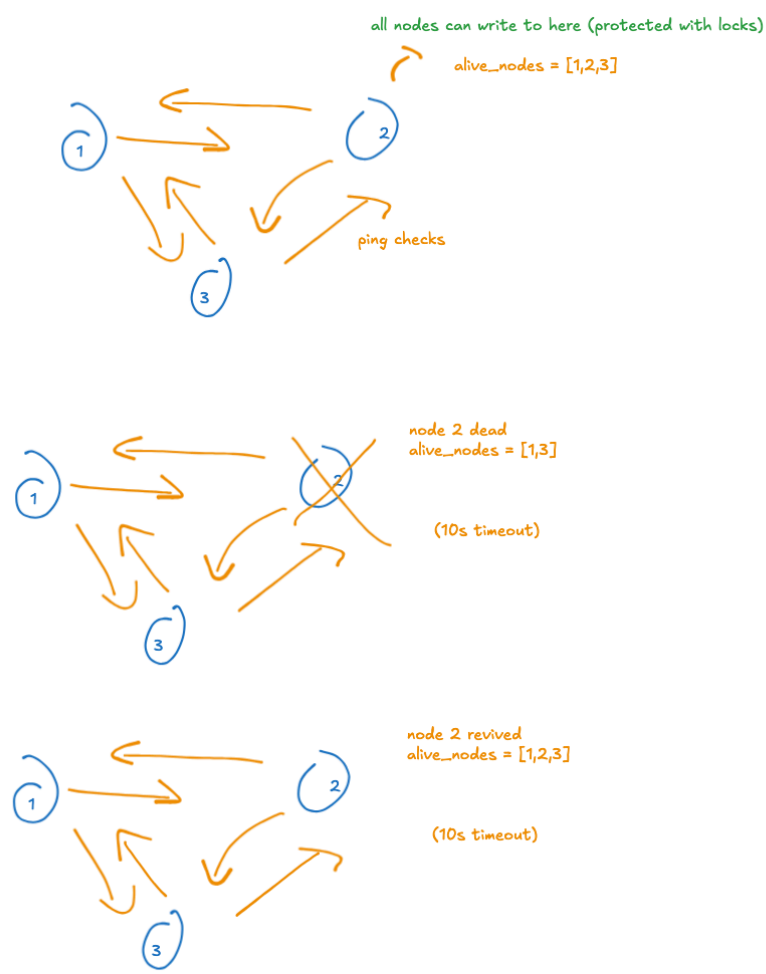
Runs in the background and pings every peer every 10 seconds. If a node stops responding it gets pulled from the active peer list, and if it comes back it gets added back in.

Docker Compose

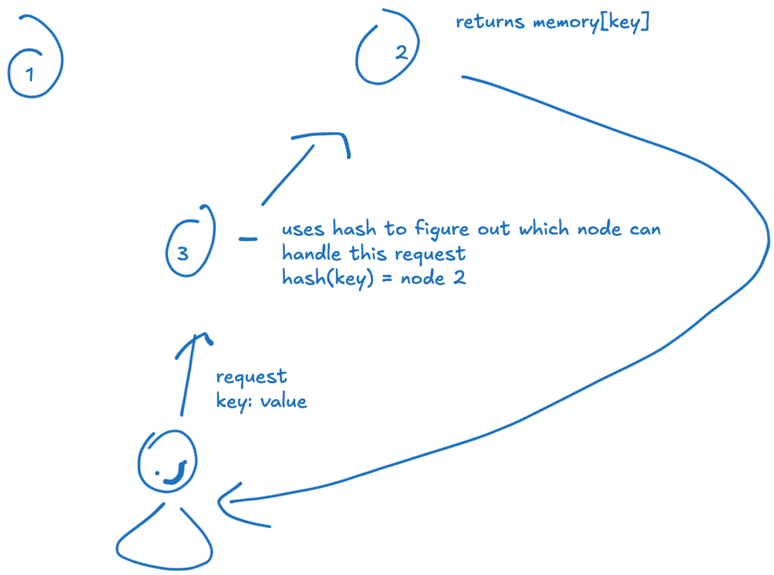
Spins up all 3 nodes on a shared internal network and exposes them on ports 5001, 5002, and 5003. Each node knows about the others through environment variables in the docker compose.

Screenshots of Design and Outputs

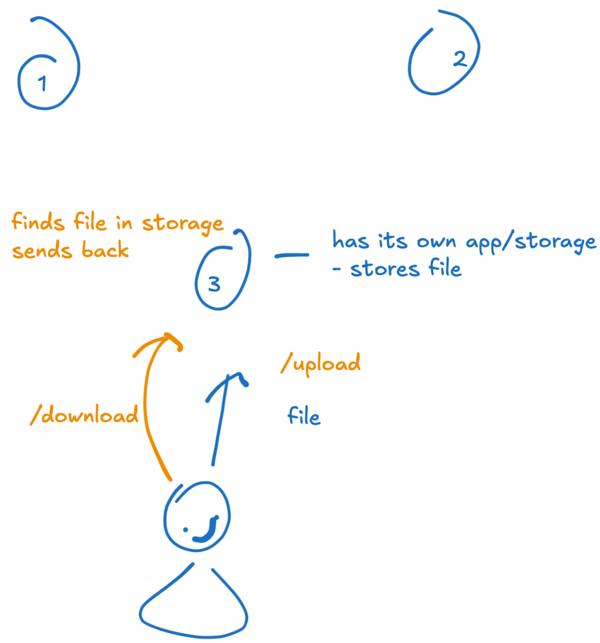
EXTRA CREDIT: Failure handle and node rejoin



KV + DHT routing



File Uploading



Individual Contributions

Ryan Tran: Video, System Design

Carlos Aguiler: Report, Docs